



Database Management Systems - Architecture

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Database System Terminologies

➤ Database Schema:

- ❖ The overall design of the database is called as database schema.
- ❖ There are two types of schema namely, physical schema and logical schema.



Physical schema

- The **physical schema** describes the database design at the physical level.
- This schema pertains to the actual storage of data and its form of storage like files, indices, etc.
- It defines how the data will be stored in a secondary storage.

Logical schema

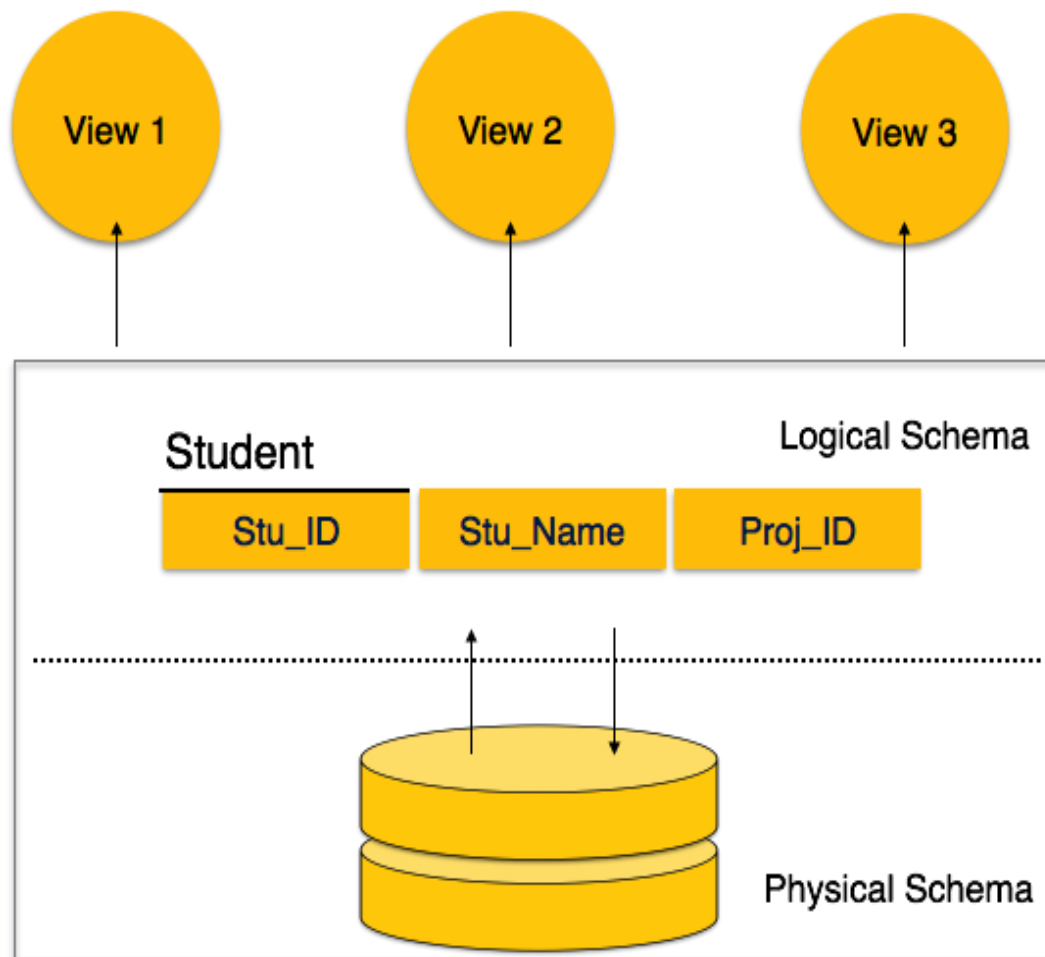
- The **logical schema** describes the database design at the logical level.
- This schema defines all the logical constraints that need to be applied on the data stored.
- It defines tables, views, and integrity constraints.



Subschema

- A database may also have several schemas at the view level, called as **subschemas**.
- It describes different views of the database.

Database Schema





Database System Terminologies

► Tables:

- The view that displays the database as a combinations of rows (records) and columns (fields).
- The first row of a table is reserved for the field names.

ID	name	dept_name	salary
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
58583	Califieri	History	62000
83821	Brandt	Comp. Sci.	92000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

(a) The *instructor* table

Database System Terminologies

➤ Field names/Column names:

- ❖ It identifies the different categories in a database.
- ❖ Top row is reserved for field names.

➤ Rows:

- ❖ It is also called a tuple or record.
- ❖ It represents a single, implicitly structured data item in a table.

Database System Terminologies

► Columns:

- ❖ In a relational database, a column is a set of value of a particular type.
- ❖ It can also be referred as attributes/fields.

	Column 1	Column 2	Column 3
Row 1			
Row 2			
Row 3			
Row 4			
Row 5			
...			

Database Architecture

- It provides the single picture of the various components of a database system and the connections among them.
- Most users of the database system are not present at the site of the database systems, but they connect it through the network.



Database Architecture

- ▶ We can differentiate client machines, on which remote database users work, and the server machines on which the database system runs.
- ▶ The top part refers to the various users of the database environment and interfaces.
- ▶ The lower part shows the internals of the DBMS responsible for storage of data and processing of transactions.

Database Architecture

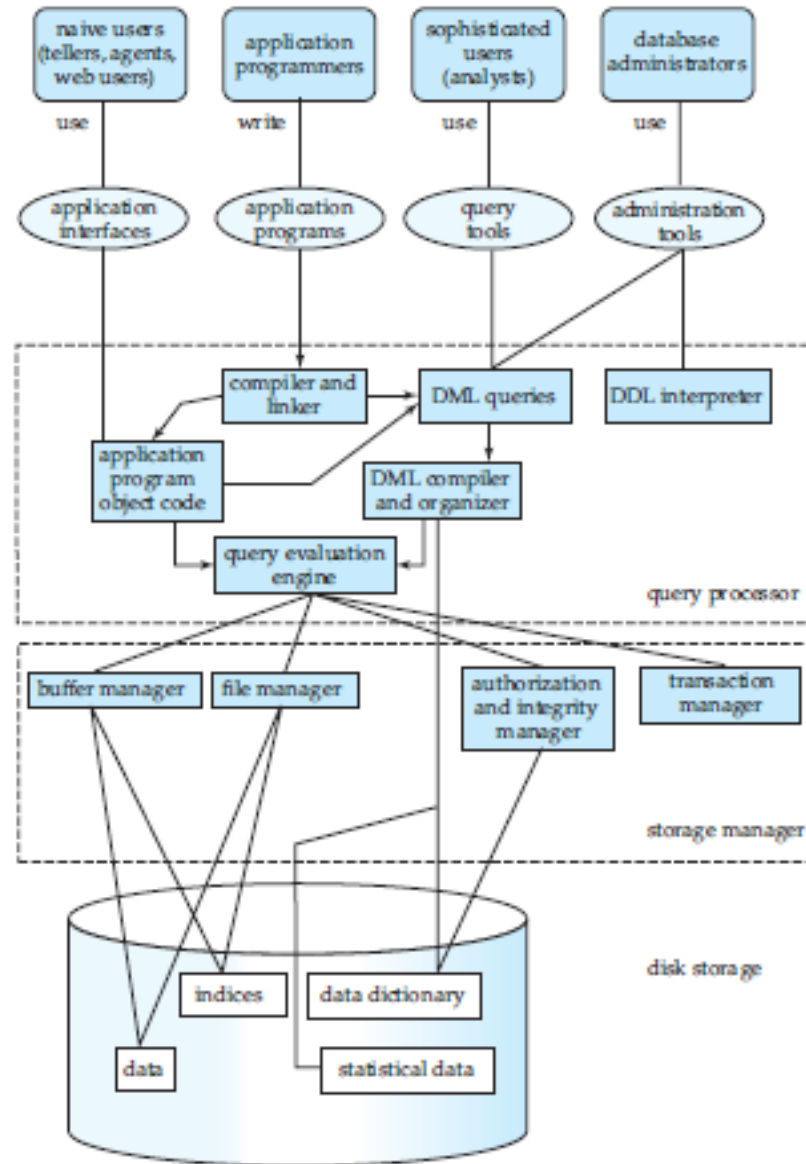


Figure 1.5 System structure.



Database Architecture

- A database system is partitioned into modules that deal with each of the responsibilities of the overall system.
- The functional components of a database system can be broadly divided into the storage manager and the query processor components.



Data Storage and Querying

- The storage manager is important as databases requires a large amount of storage space.
- Corporate databases range in size from hundreds of gigabytes to terabytes of data.
- As main memory of computers cannot store much information, these information is stored on disks.
- Data are moved between disk storage and main memory as needed.



Data Storage and Querying

- Movement of data to and from disk is slow compared to CPU.
- The query processor is important because it helps the database system to simplify and facilitate access to data.
- The query processor allows database users to obtain good performance while working at the view level.



Storage Manager

- The storage manager is the component of a database system that provides the interface between the low-level data stored in the database and the application programs and queries submitted to the system.
- It is responsible for storing, retrieving, and updating data in the database.

Components of storage manager

➤ Authorization and integrity manager

It tests for the satisfaction of integrity constraints and checks the authority of users to access data.

➤ Transaction manager

It ensures that the database remains in the consistent state despite system failures.



Components of storage manager

➤ File manager

It manages the allocation of space on disk storage and the data structures used to represent information stored on disk.

➤ Buffer manager

It is responsible for fetching data from disk storage to main memory.

Storage Manager

- The storage manager implements several data structures as part of the physical system implementation.
- **Data files**
It stores the database itself.
- **Data dictionary**
It stores the metadata about the structure of the database.
- **Indices**
It can provide fast access to data items.

Query Processor

- The query processor components include

- **DDL interpreter**

It interprets DDL statements and records the definition in the data dictionary.

- **DML compiler**

It translates DML into an evaluation plan which is understood by the evaluation engine.

- **Query evaluation engine**

It executes low level instructions generated by the DML compiler.